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Planning of carbon capture storage deployment using process graph approach

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ABSTRACT

Carbon capture and storage (CCS) is an emerging technology to mitigate carbon dioxide (CO_2) emissions from industrial sources such as power plants. However, retrofitting a power plant for carbon capture causes an increase in unit power cost due to parasitic power losses as well as capital outlays for additional process equipment. Mathematical optimisation and pinch analysis techniques have been used to systematically plan for the retrofit of power plants. In this work, the planning of power plants retrofit along with CO_2 source-sink matching is analysed using *process graph* (P-graph) optimisation technique. CO_2 sources are assumed to be characterised by fixed flowrates and operating lives; while CO_2 sinks are characterised by storage capacity limits and earliest time of availability. Illustrative case studies are solved to demonstrate the approach.

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1. Introduction

Global carbon dioxide (CO_2) emissions are now regarded as a major issue to society. Climate change is considered as a critical problem, with the current atmospheric CO_2 concentration now in excess of safe limits [1]. In addition, human activities continue to add a steady stream of greenhouse gases to the atmosphere. Power generation from fossil fuels (coal, oil and natural gas) contributes to a significant portion of these CO_2 emissions. Fossil fuels currently supply more than 85% of the energy used worldwide due to their low cost, availability, reliability, and energy density [2,3]. In order to reduce the climatic impacts, efficiency improvement on current technologies, fuels substitution and utilisation of low-carbon energy for cleaner electricity generation have been implemented [4,5]. However, fossil fuels will probably remain as a major contributor to the world's power generation mix in the future, due to the limitations of many low-carbon alternatives [6,7]. In particular, fossil fuels continue to dominate the energy markets, especially in developing countries characterised by growing economies and rising energy demands. In addition, most renewable energy options are often subject to significant geographic limitations. Although nuclear power is a mature, low-carbon alternative to fossil fuels, it has recently raised worldwide concerns on environmental and safety issues after several major accidents, e.g. the 2011 Fukushima disaster in Japan. These factors contribute to the requirement for the deployment of *carbon capture and storage* (CCS) technology in order to mitigate climate change by reducing industrial CO_2 emissions.

As its name suggests, CCS first entails *carbon capture* (i.e., isolation of CO_2 from combustion flue gas) and then *carbon storage* (i.e., disposal of the CO_2 in an appropriate geological storage reservoir). Current capture technologies include oxy-fuel combustion (OFC), chemical looping combustion, pre-combustion using integrated gasification combined cycles (IGCC), or post-combustion capture via flue gas scrubbing (FGS) [8–11]. Typically, 80–90% of CO_2 from power plant exhaust gases can be captured using these technologies and subsequently, compressed for secure storage in

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various geological formations, such as depleted oil or gas reservoirs, inaccessible coal deposits, saline aquifers and other geological structures of sufficient integrity.

Note however that, retrofitting power plant with CCS has some major issues, one of which is the energy consumption of additional process equipment to isolate and compress CO_2. When plants are retrofitted for capture, the power output of the retrofitted plant is 15–20% lower than original output due to parasitic energy demands (i.e. the additional energy demands for CO_2 capture) [9]. In addition, capital cost for plants with CCS will be 25–50% higher than that of baseline plants due to the additional process equipment, such as air separation units for OFC and absorption columns for FGS [9]. Subsequently, these equipment cause a drop in plant thermal efficiency of 5–10%, resulting in an increase in the cost of electricity generated [12]. To compensate for the power loss, additional electricity also needs to be generated from new plants in order to maintain the grid-wide power output prior to CCS deployment, which ultimately contributes to incremental costs. Alternatively, electricity may need to be imported from nearby regions (if these have surplus supply); such measures may, of course, potentially compromise energy security or independence of a country. Failure to compensate for the energy penalties incurred by CCS will result in power shortages. All of these complex considerations result in the need for proper planning of CCS deployment in power generation sector. Market allocation optimisation model has been used to analyse economic aspect of CCS systems [13]. Life cycle assessment has also been used to study the trade-offs between different environmental impacts after implementation of CCS systems [14]. However, there are still significant uncertainties with respect to economics of CCS systems and technical uncertainties in CO_2 life cycle [15], mainly are grid power problems and CO_2 source-sink matching.

Pinch analysis was first introduced to address CCS planning problem, particularly for carbon capture planning [16–18]. In the seminal work of Tan et al. [16], which was an extension based on carbon emission pinch analysis [19], useful insights and performance targets (e.g. minimum extent of retrofit) to facilitate the CCS retrofit planning stage are obtained using the graphical tool known as the *CCS planning composite curve*. However, there are several limitations in this approach. First, it only handles highly aggregated energy sources and demands, in which planning can only be made at the sectorial level. Besides, various design constraints and economic considerations for detailed planning cannot be included. Shenoy and Shenoy [17] employed *table algorithm* and *nearest neighbour algorithm* to design the carbon emission networks, and followed by total cost optimisation using mixed integer linear programming (MILP) formulation. A significant advantage of this methodology is that network can be designed separately, without requiring the cost data. Therefore, many alternative networks can be obtained by just varying order of satisfaction in table algorithm. The drawback of this methodology is the optimum network can only be obtained after going through few distinct stages of analysis. A recent work of Sahu et al. [18] improves the previous work of Tan et al. [16], where compensatory power was assumed to be generated only from carbon-neutral sources. The group makes use of algebraic technique to handle cases where compensatory power is generated from both carbon-neutral and non-carbon-neutral sources.

In order to overcome the limitation of pinch analysis techniques, several works based on *mathematical optimisation* techniques have been developed for the planning of CCS deployment. These include those based on superstructure model [20–22] and automated targeting technique [23,24]. Mathematical optimisation technique is preferable, when detailed planning scenarios are encountered. Such models also provide an opportunity to integrate more complex, case-specific goal functions especially to handle the deployment of CCS retrofit with concern for cost-effectiveness [25].

On the other hand, different techniques have also been developed for carbon storage planning problem, in order to match multiple CO_2 sources and sinks (storage sites). Graphical techniques based on pinch analysis were developed to cater for capacity [26] and injectivity constraints [27]. Diamante et al. [28] improved their previous work [27] by considering time availability of sources and sinks, simultaneously with injectivity limits. However, geographical distances and pipeline costs between various sources and sinks are neglected due to inherent simplifications and lower expandability of pinch approaches.

Different mathematical optimisation models have also been presented for the carbon storage problem. A discrete-time MILP model was developed for optimal source-sink matching with temporal, injection rate and storage capacity constraints [25]. This constraint is tackled by dividing a finite planning horizon into discrete time intervals. However, this model is only suitable for mid-term planning of CCS option for plants located in close geographical proximity to sinks. For an increase in precision, shorter time intervals are required in the model, which results in the increase in model variables with the associated penalties in computational effort. A related continuous-time MILP model for CO_2 source-sink matching in CCS systems also developed [29]. This model accounts for CO_2 emission penalties result from generating electricity to compensate for grid-wide CCS power losses [16,17,20,21]. The main assumption of Tan et al. [29] is by omitting injection rate, since physical characteristic of CO_2 sinks capacity is more significant. Later, Lee and Chen [30] proposed an improved MILP model for similar problems. Lee et al. [31] has recently presented a unified multi-period MILP model to consider CCS retrofit planning and CO_2 source-sink matching simultaneously.

As mentioned, different approaches for planning CCS deployment have been proposed, each with their own unique advantages and disadvantages. In this work, we propose a novel alternative approach to CCS system planning based on *process network synthesis* (PNS), which is based on *process graph* (P-graph). P-graph methodology is a powerful approach which utilises graph theory to perform an efficient search of the solution space of a given problem domain. P-graph framework was first introduced by Friedler et al. [32] for synthesis of process system. This approach resorts to the well-established mathematics of graph theory and it is heavily based on a unique class of graphs as well as combinatorial techniques [32]. It focuses on structures of the whole system and rigorously examines all possible structures from mathematical perspective, while allowing for a more efficient search of the solution space than is possible from MILP approaches. A wide range of successful application has then been reported, which include molecular design [33], reaction pathway synthesis [34,35], synthesis of separation network [36–39], heat exchanger network synthesis [40], process synthesis [41–43], and energy supply chain [44,45]. However, no attempt has been reported for the use of P-graph for CCS planning problem, which is the main aim of this work.

This paper proposes a P-graph approach for the systematic planning of CCS deployment in the power generation sector. The rest of the work is organised as below. First, a formal problem statement is given. A brief explanation on methodology used in this paper is next discussed. A literature case study on carbon capture is adapted as base case and solved using P-graph approach. Furthermore, different parameters are used to illustrate the potential future scenarios in CCS planning; and sensitivity analysis is carried out by generating Pareto optimal curve. Extensions are then developed from this base case to determine appropriate source-sink matching for carbon capture planning based on temporal

and storage capacity limit. Finally, conclusions and prospects for future work are given at the end of this paper.

2. Problem statement

The main objective of the problem is to minimise the carbon footprint of the entire power sector, by selecting power plants to be retrofitted and by identifying the best technique for retrofitting a given plant, while keeping the incremental cost of electricity within a specified limit. A superstructure representation of such a network is presented in Fig. 1, showing all possible connections between the sources-technology-sinks.

The specific problem to be addressed is stated formally as follows:

1. The carbon capture system is assumed to be comprised of m CO_2 sources (i.e., fossil fuel fired power plants) and n CO_2 sinks (i.e., storage reservoirs). The choice of carbon capture (CC) technology, j ($j = 1, 2, \ldots, t$) for each plant is limited to a maximum of one and each capture technology has a fixed efficiency.
2. Each CO_2 source i ($i = 1, 2, \ldots, m$) is characterised by fixed captured CO_2 flowrate that corresponds to the maximum removal from the plant's flue gas. This flowrate represents the maximum amount of CO_2 that can be captured from the source if the decision is made to retrofit; on the other hand, CO_2 is merely released to the atmosphere if there is no implementation of CO_2 capture. The operating life of each source i is defined.
3. Each CO_2 sink k ($k = 1, 2, \ldots, n$) is characterised by an upper limit for CO_2 storage capacity over its lifetime, as determined by factors such as time of availability, injection rate and storage capacity limit. Each sink requires a fixed input of CO_2 (e.g. flowrate).
4. Due to large uncertainties inherent with cost estimation, an aggregate cost limit is used with parameters based simply on relative costs of power from unmodified, retrofitted and new power plants.

3. P-graph for carbon capture planning

The concept of P-graph is briefly explained in this section. P-graph is a bipartite graph for umabiguous representation of processing networks. There are two types of vertices: horizontal bars representing operating units; and solid circles representing material or energy streams; vertices are connected by directed arcs [32]. A simple conventional and P-graph representation of an operating unit (e.g. a gas turbine) is shown in Fig. 2.

There are several combinatorial instruments associated with P-graphs [46]. First, the *axioms* and *theorems* are to ensure representation unabiguity and consistencies of the resulting superstructures and solution networks. The other instruments are the three main algorithms which include superstructure construction. *Maximal structure generation* (MSG) first determines an overall structure for the problem which encompasses all possible solutions. Next, *solution-structure generation* (SSG) enables multiple candidate solutions to be identified as subsets of the maximal structure. Finally, *superstructure optimisation* is achieved via *accelerated branch and bound* (ABB) to identify the best solution among the various candidates based on specified criterion [45,46]. It should be noted that ABB allows for a more efficient search of solution topologies than is possible with conventional branch-and-bound solvers used for MILP models; thus, in the case of large problems, it is possible to achieve significant reductions in solution time. Also, the P-graph approach is also able to identify near-

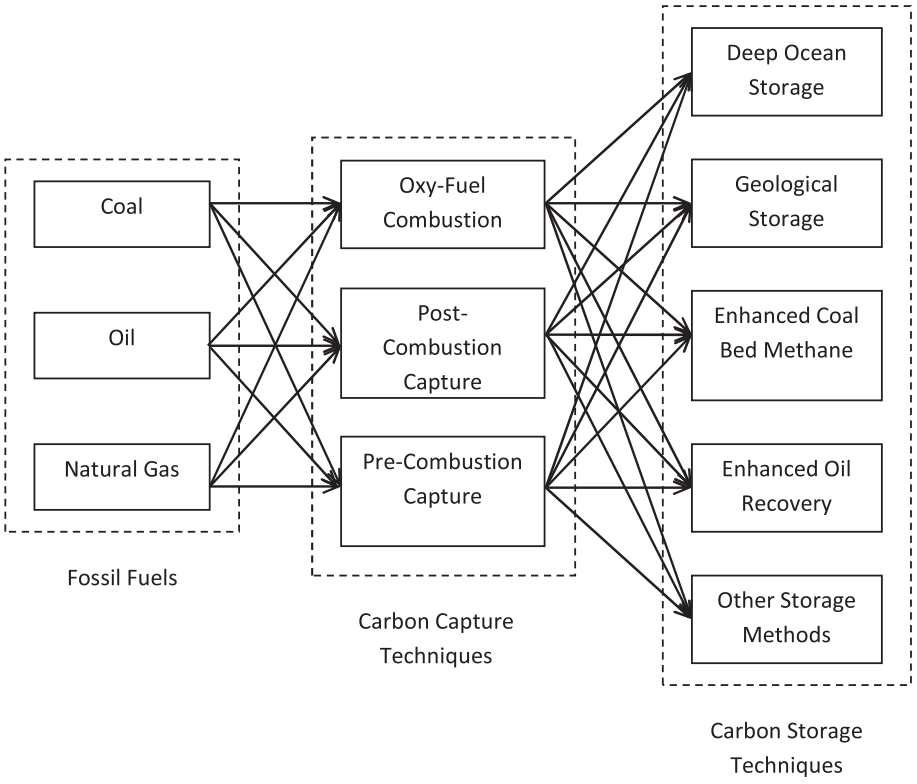


Fig. 1. Schematic representation for CCS retrofit.

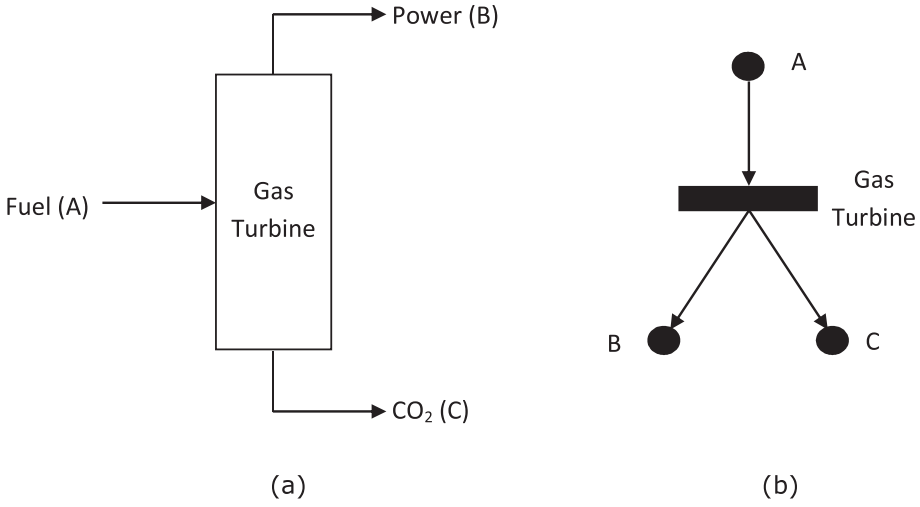


Fig. 2. A power plant representation in (a) conventional process flow diagram, and (b) P-graph.

optimal solutions (in addition to the global optimum), a feature which is not possible using other mathematical optimisation techniques, but which may prove valuable for practical decision-making. By comparison, the MILP approach will yield the global optimum, but will not be able to identify comparable solutions.

To represent carbon capture options for power plants (i.e. first part of the superstructure in Fig. 1) on P-graph, the following steps are needed:

a. Identification of materials and streams
b. Identification of candidate operating units
c. Specification of units' performance and investment cost
d. Identification of upper and lower bounds of materials, streams and operating units

Firstly, the streams involved in this superstructure were identified as power generated and the power generation cost, as listed in Table 1. Each power plant generates power at different capacity with a specific cost; therefore power and cost are the input streams in P-graph. When a carbon capture technique is chosen, the power output after retrofit, power loss and compensatory power are then determined, as well as its associated costs due to retrofit and compensatory power. These are intermediate streams in P-graph representation. The final outputs in P-graph representation are the total power output from all power plants (including compensatory plants), and the overall cost from all plants.

In P-graph, operating units are capable of transforming certain materials or streams into other ones in order to produce desired products from the specified raw materials through the defined intermediates [43]. In the second step, any possible candidate operating units are listed, as shown in Table 2. All candidate operating units feature different performance and overall costs. Therefore, in the third step, these will be specified for the solver to choose the optimal operating units based on the objective. The performance of the units is specified by the amounts of outputs per unit of chosen inputs; while overall costs of operating units are related to the operating costs and investment costs. In this case, the performance of each unit is given in Section 4 and the overall costs are defined by the CO_2 emitted by the operating units.

Finally, the upper and lower bounds for all streams and operating units are specified. The limitation is usually depending on the operating unit capacities or availability of resources. This step is important for the solver to choose optimal units and streams to be used, starting with the most profitable and efficient options [43]. For this case, the lower bounds for power plants *i* for all technology used are the respective initial output capacities. This is to make sure

Table 1
Streams for P-graph model.

Symbols	P-graph classification	Description
$Power_in_i$	Input	Initial power capacity for each plant
$Cost_in_i$	Input	Cost corresponding to power generated from each plant
OFC_PL_i	Intermediate	Power loss from OFC for each plant
OFC_CC_i	Intermediate	Cost from compensatory plant for OFC in each plant
OFC_PC_i	Intermediate	Power from compensatory plant for OFC in each plant
OFC_RP_i	Intermediate	Retrofitted power output from OFC from each plant
OFC_RC_i	Intermediate	Cost output from OFC from each plant
FGS_PL_i	Intermediate	Power loss from FGS for each plant
FGS_CC_i	Intermediate	Cost from compensatory plant for FGS in each plant
FGS_PC_i	Intermediate	Power from compensatory plant for FGS in each plant
FGS_RP_i	Intermediate	Retrofitted power output from FGS from each plant
FGS_RC_i	Intermediate	Cost output from FGS from each plant
$IGCC_PL_i$	Intermediate	Power loss from IGCC for each plant
$IGCC_CC_i$	Intermediate	Cost from compensatory plant for IGCC in each plant
$IGCC_PC_i$	Intermediate	Power from compensatory plant for IGCC in each plant
$IGCC_RP_i$	Intermediate	Retrofitted power output from IGCC from each plant
$IGCC_RC_i$	Intermediate	Cost output from IGCC from each plant
Power	Output	Total power output from all plants
Cost	Output	Overall cost of electricity from all plants

Table 2
Operating units for P-graph model.

Symbols	P-graph classification	Description
OFC_i	Operating unit	Retrofit with OFC technology
FGS_i	Operating unit	Retrofit with FGS technology
$IGCC_i$	Operating unit	Retrofit with IGCC technology
NR_i	Operating unit	Non Retrofit
OFC_Comp_i	Operating unit	Compensatory power plant for plant with OFC
OFC_Sum_i	Operating unit	Summation of output for plant using OFC
FGS_Comp_i	Operating unit	Compensatory power plant for plant with FGS
FGS_Sum_i	Operating unit	Summation of output for plant using FGS
$IGCC_Comp_i$	Operating unit	Compensatory power plant for plant with IGCC
$IGCC_Sum_i$	Operating unit	Summation of output for plant using IGCC

only one technology is chosen for each power plant. Upper bound is not required in this case because there is no limitation defined for the operating unit capacities or resources availability.

The P-graph model consists of three parts, starting from power to be generated by each plant and their corresponding costs, followed by technologies, and finally the total final power output and corresponding costs. For the first part of P-graph model, the graphical representation for using carbon capture options of OFC, FGS and IGCC is shown in Fig. 3 (a). Power targeted for each plant is connected to different retrofit technology. At the same time, total power cost is proportional to the power generated. CO_2 released will be determined by power targeted for each power plant and fuel used. Relationships of power generated and power cost to retrofitt technology are given in Equations (1) and (2); both equations show simple balance of power and power cost.

$$P_i = \sum_j P_{ij} \quad \forall i \tag{1}$$

$$C_i = \sum_j P_{ij} c \quad \forall i \tag{2}$$

The second part of the P-graph model is shown in Fig. 3(b), which is between power generated and the associated total power cost with the retrofitting technology. Note that Fig. 3(b) only shows for one the carbon capture technique of OFC; the other carbon capture techniques will take the same form. Equations (3) and (4) show these relationships mathematically. For both equations, the first terms in the equations give the power and cost after implementing a carbon capture technology; while the second term gives the power or cost resulting from the new compensatory plant.

$$P_j = P_{ij} x_{ij} (1 - L_j) + P_{ij} x_{ij} L_j \quad \forall j \tag{3}$$

$$C_j = P_{ij} x_{ij} (1 - L_j) A_j + P_{ij} x_{ij} L_j B \quad \forall j \tag{4}$$

The choice of CCS technologies for each plant is limited to maximum of one. When a retrofitting technology is chosen, the binary variable x_{ij} is denoted as one, or zero otherwise.

$$\sum_j x_{ij} \leq 1 \quad \forall i \tag{5}$$

$$x_{ij} \in \{0, 1\} \quad \forall i, j \tag{6}$$

When the plant is chosen to be retrofitted, power loss (L_j) follows the value given in each case study. If no modification of power plant is chosen, L_j is equal to zero. Cost coefficients (A_j and B) are dimensionless, and give the relative cost of electricity in comparison with the baseline cost.

The final output is equal to the summation of all outputs from all technologies, as shown in the Equations (7) and (8) below. Fig. 3(c) shows the P-graph representation of these relationships.

$$P_{\text{total}} = \sum_j P_j \tag{7}$$

$$C_{\text{total}} = \sum_j C_j \tag{8}$$

The objective function of all problems studied in this paper is to maximise the reduction of total carbon emissions:

$$\max \sum_i P_i F_i \left[\sum_j RR_j x_{ij} \right] - D \left[\sum_i P_i \left(\sum_j L_j x_{ij} \right) \right] \tag{9}$$

The first term gives the total reduction in emissions from carbon capture retrofitting, while the second term gives the additional emissions from new plants that are added to compensate the parasitic power losses. Hence, the summation of two terms gives the emissions reduction relative to the baseline level.

In order to carry out CCS planning using P-graph, data is to be specified for power plants (i.e., rated capacities and emission factors) and carbon capture technologies (i.e., removal ratios and power losses). P-graph methodology will then select the right power plants for retrofit with appropriate carbon capture technology to fulfil CO_2 reduction targets, cost limits and power losses. In this work, the P-graph software PNS Studio [47] is used to implement the model using a PC with a 2.67 GHz processor and 2 GB of RAM. The methodology is illustrated by case studies that follow.

4. Base case

A case study from Tan et al. [21] is adapted for use here. As shown in Table 3, six power plants are to be considered for CCS retrofit. Each of these plants has specified capacities and emissions factors based on its fuel. The total power generated in this region is 2250 MW and CO_2 produced corresponding to this power

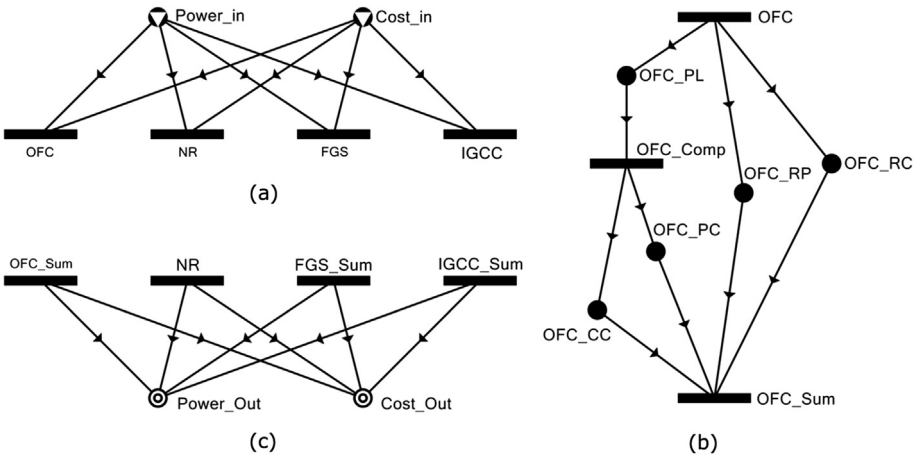


Fig. 3. P-graph model for different carbon capture options: (a) power and cost for different retrofit technologies; (b) input and output for a retrofit technology (OFC); (c) total power output and cost from all retrofit technologies.

Table 3
Plant data for Base Case.

Plant	Fuel	Emission factor (t CO_2/MWh)	Power (MW)	Emission (t CO_2/h)
1	Coal	1	600	600
2	Coal	1	500	500
3	Natural gas	0.5	250	125
4	Natural gas	0.5	300	150
5	Natural gas	0.5	400	200
6	Oil	0.7	200	140
Total			2250	1715

generation is 1715 t/h; this is equivalent to average carbon footprint of 0.762 t/MWh. In this case study, two carbon capture technologies, i.e. FGS and OFC are considered as choices for reducing carbon emission. Note that a technological compatibility restriction is assumed here, i.e. plant 3 is not suitable for OFC retrofit. OFC is more difficult to be implemented in a natural gas fired gas turbine power plant; OFC retrofit is more suited to Rankine cycle plants [48]. The removal ratio (RR_j) and relative power loss (L_j) of FGS are 0.8 and 0.2; while those parameters for OFC are 0.9 and 0.25 respectively [21]. Power loss results from retrofitting will be compensated, and emission factor of compensatory plant (D) is taken as 0.1 t CO_2/MWh. P-graph representation of the case study is given in Fig. 4.

For economic analysis, it is assumed that electricity produced in the retrofitted plants costs 60% more expensive than that from unmodified plants. The electricity generated from compensatory plant is assumed to be 40% more expensive than the baseline cost (i.e., in practice the compensatory power must also be low-carbon in order to justify CCS deployment). Hence, the relative cost coefficients (A_1, A_2, A_3 and B) used for Equation (4) are given as 1, 1.6, 1.6 and 1.4 [21]. The main objective is to minimise carbon footprint without increasing the overall electricity cost by more than 30% than the base cost. This is then solved in PNS Studio using SSG and LP algorithms.

The optimal result shows that Plants 1 and 2 requires CCS retrofit with OFC, while the rest of the plants remain unmodified. The summary of the results is shown in Table 4. Final carbon emission after CCS retrofit is determined as 752.5 t/h as reported in the last entry of column 4. With total power generation of 2250 MW, this translates into a carbon footprint of 0.334 t/MWh. This indicates the greatest possible reduction of carbon footprint is about 0.428 t/MWh, which correspond to 56.12% decrement from baseline level of 0.762 t/MWh. The resulting electricity cost is 26.89% more expensive than the baseline cost, which is within cost constraint of 30%. The final contributions to the total power output are 1150 MW (51.1%) from unmodified plants, 825 MW (36.7%) from retrofitted plants and 275 MW (12.2%) from new compensatory plant. The overall results are further summarised in the second column of Table 5. These results match with the results when the case study is solved using the mathematical model of Tan et al. (2010) [21]. The P-graph representation of the solution for Base Case is shown as Fig. S1 in a supplementary file.

5. Scenario analysis

In previous section, the Base Case Study shows the power generation scheme with various assumptions. In this section, several scenarios are analysed to observe their consequences on CCS retrofit.

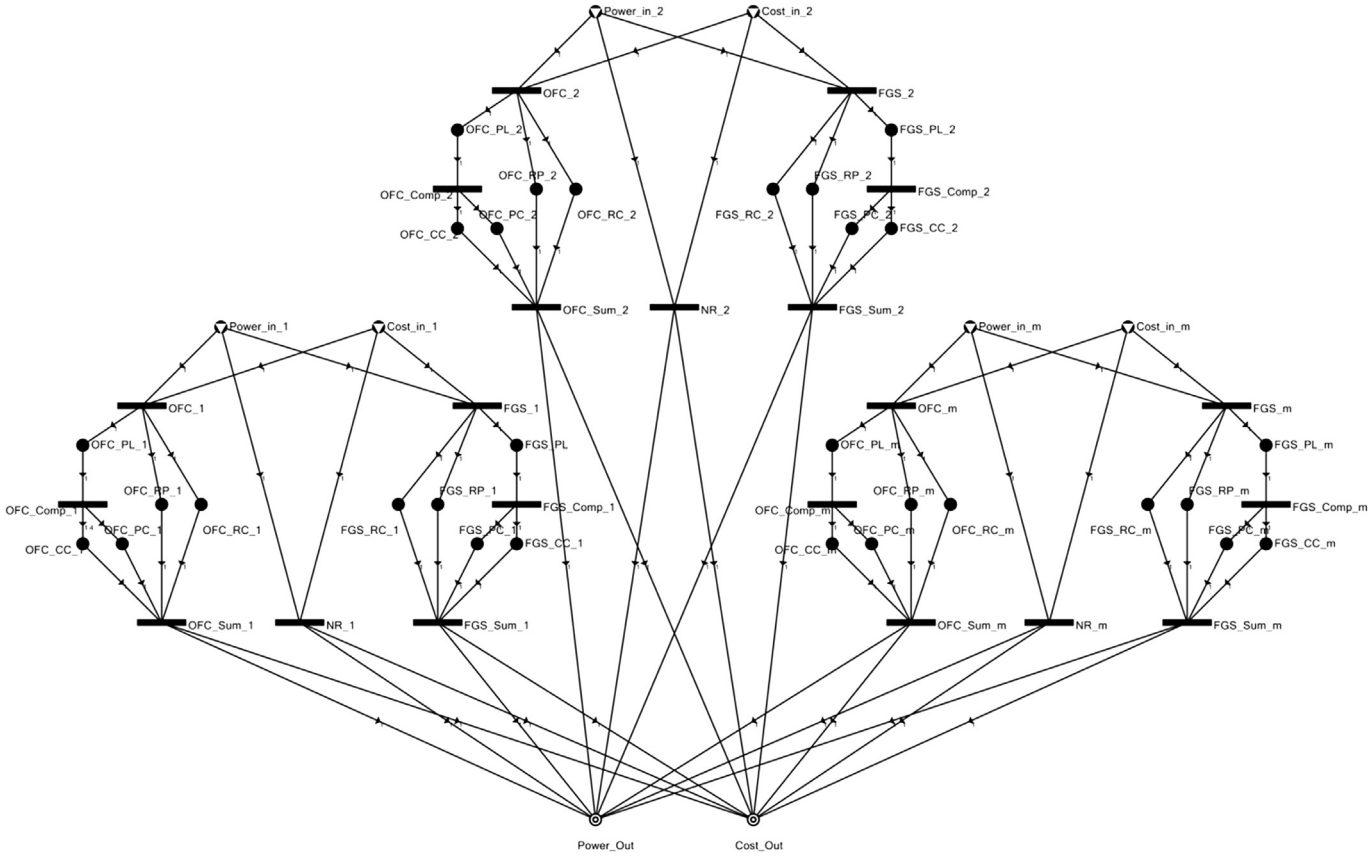


Fig. 4. P-graph representations for case study (Base Case).

Table 4
Power sector data after CC retrofit in Base Case.

Plant	CC technique	Final power output (MW)	Final carbon emissions (t CO_2/h)	Increase in cost per unit of power (%)
1	OFC	450	60	60
2	OFC	375	50	60
3	None	250	125	0
4	None	300	150	0
5	None	400	200	0
6	None	200	140	0
Compensatory	None	275	27.5	40
Total		2250	752.5	26.9

Table 5
Key results for Base Case and Scenarios 1–3.

Parameter	Base case	Scenario 1	Scenario 2	Scenario 3
Final carbon footprint (t/MWh)	0.334	0.278	0.295	0.308
Relative reduction (%)	56.12	63.56	59.90	59.65
Grid relative power cost	1.269	1.296	1.296	1.274
Power from unmodified plants (MW)	1150	850	1200	1150
Power from retrofitted plants (MW)	825	1050	1050	880
Power from new plants (MW)	275	350	350	220

5.1. Scenario 1

In the first scenario, it is assumed that all power plants implementing CCS will be subsidised by government, thus reducing the power cost from retrofitted power plants. It was reported that major policy initiatives are required for reducing the carbon emissions globally and subsidies to low-carbon technologies such as CCS is an important element of future climate policies around the world [49].

This scenario makes use of the same data as in the Base Case, except that the increment cost of electricity in the retrofitted plant will be assumed as 50%, as compared to the initial assumption of 60% (as compared to unmodified plants). It is assumed that 10% increment is to be subsidised by the government. Having financial support from the government, it is expected that more power plants will undergo CCS retrofit, and hence lower total CO_2 emission. The main objective remains the same as that in Base Case, which is to maximise CO_2 reduction while keeping electricity cost increment within 30% from base cost.

Solving the case study using SSG and LP algorithms in PNS Studio, yields an optimal result where Plants 1, 2 and 4 require OFC retrofit, while keeping the others unchanged. The final carbon emission after CCS retrofit is determined as 625 t/h. The summary of this scenario is given in the third column in Table 5. The final carbon footprint of resulting electricity mix is 0.2778 t CO_2/MWh, which is 63.56% lower than the baseline level. The increment of electricity cost is still within 30% of the base cost. The overall optimum solution to this scenario is shown using P-graph representation, which is included as Fig. S2 in the supplementary file.

As compared to the Base Case, this results in higher reduction in carbon emissions and slightly higher unit cost of power. It shows one extra power plant can install OFC compared to Base Case without compromising the economic limit set beforehand. It is obvious that having subsidies support, the power sector is able to have more plants retrofitted and reduce emission of greenhouse gases. This result, when compared with that of Base Case, clearly indicates how sensitive are the carbon footprint reduction to the cost increment from retrofitting a power plant.

5.2. Scenario 2

In Base Case and Scenario 1, the power demand is assumed to be same and so does the power output from each plant. Power demand is estimated to be about 30% higher in 30 years time, as economic output grows more than doubles and prosperity expands across a world with population growing rapidly [50]. The trend is especially pronounced in the developing world, where many countries are characterised by rapid economic development, population growth and rising energy intensity. Therefore power output of some power plants shall be increased to meet the demand. In this scenario, power output of Plants 3, 4 and 6 are increased by 100 MW, 150 MW and 100 MW respectively. Increasing the power output will increase the carbon emissions proportionately as well. The emission factors are kept constant in this scenario, thus the carbon emissions of Plants 3, 4 and 5 are now 175 t/h, 225 t/h and 210 t/h. Apart from these, the other data are remained the same as Base Case.

This scenario is solved using SSG and LP algorithms in PNS Studio. The results of this scenario is summarised in forth column of Table 5. The optimal solution is to retrofit Plant 1, 2 and 6 using OFC and keeping the others unmodified. The final carbon footprint is 0.2946 t/MWh; this is lower than Base Case but higher than Scenario 1. The corresponding reduction of carbon footprint is only 59.90% of the baseline level while the cost increment is kept within 30% of base cost. The carbon footprint reduction is still lower than that for Scenario 1, but the cost increment is the highest among all cases. The graphical representation of optimum solution for retrofitting power plants in this scenario is also included as Fig. S3 in the supplementary file.

Similar to Scenario 1, one extra power plant is chosen for CCS retrofit as compared to Base Case. However, this scenario still has higher carbon footprint than that of Scenario 1; while both Scenarios 1 and 2 have the same cost increment. This indicates that increment in power demand does not help in carbon emission reduction as much as having subsidies from government when CCS is employed.

5.3. Scenario 3

Carbon capture technologies are still relatively immature (as compared to other clean fuel technologies) and there is potential in improving them further, for example through effective use of heat integration within power plants [51]. In this scenario, it is assumed that the removal ratios of carbon capture technologies increase while the relative power losses decrease. Same data as Base Case is used here, except that the removal ratio of both carbon capture technologies and their power losses, which are given in Table 6. This scenario is also solved using SSG and LP algorithms in PNS Studio.

Solving this scenario yields the optimal results in which Plants 1 and 2 were retrofitted with OFC technology, i.e. identical to that of the Base Case. A final carbon footprint of 0.3076 t/MWh is achieved and corresponds to a reduction of 59.65% from the baseline level. The unit power cost increases by 27.4%, which is slightly higher than that of Base Case. The summary is given in the fifth column in Table 5. The graphical representation of the optimum solution for

Table 6
Technology new data for Scenario 3.[8,51].

Technology	Removal ratio	Relative power losses
OFC	0.95	0.20
FGS	0.90	0.10

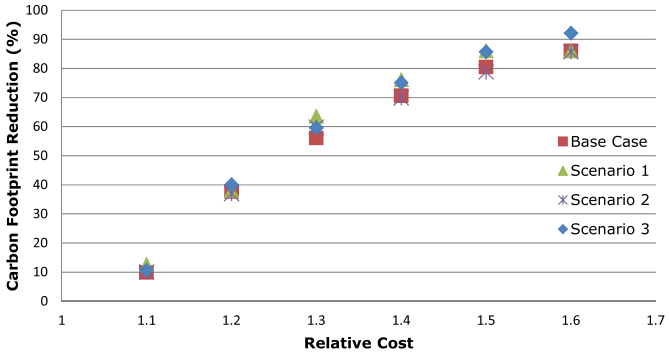


Fig. 5. Carbon footprint and cost trade-offs for Base Case and Scenario 1–3.

Table 7
CO_2 source and sink data for Case Study 2.

Sources	CO_2 flowrate (t/h)	Maximum CO_2 captured (Mt)	Sinks	Maximum CO_2 storage (Mt)
Plant 1	540	200	Sink 1	100
Plant 2	450	150	Sink 2	140
Total		350	Total	240

Table 8
Distance between sources and sinks for Case Study 2.

Sources	Distance (km)	
	Sink 1	Sink 2
Plant 1	200	400
Plant 2	500	250

this scenario is shown as Fig. S4 in the supplementary file. The result shows that further improvement of technology does not affect the result much, when compared with that in the Base Case. While Base Case and this scenario have chosen the same plants for modification, this scenario has higher unit power cost, indicating that improving carbon capture technology without subsidy will lead to higher unit power cost with only slightly lower carbon emissions.

5.4. Trade-off analysis

In this section, a trade-off analysis on the relationship between economic constraint and carbon emission is done by using Pareto analysis. The Pareto frontier is plotted to account for trade-off between environmental and economic goal for Base Case and Scenario 1 to 3. It is important to analyse the trade-off as the increase in cost of power generated is an inevitable outcome of the extensive retrofit. An aggregate cost limit is used based on relative costs of power from unmodified, retrofitted and new power plants due to large uncertainties associated with cost estimation. The sensitivity of the average carbon footprint of the entire power sector to the cost limit is analysed as shown in Fig. 5 for Base Case as well as the three scenarios. The points in the figure represent the frontier of Pareto-optimal solutions, where no improvement can be made to the carbon footprint without sacrificing the cost, or vice-versa.

Table 9
Source-sink matching matrix for Case Study 2 (CO_2 load in Mt).

	Sink 1	Sink 2	Storage deficit
Plant 1	100		100
Plant 2		140	10

The Pareto front in Fig. 5 shows that the reduction in carbon footprint for Base Case increases steadily as the cost limit is relaxed in the range of 1.0–1.4. Further increments in cost show diminishing returns. Carbon footprint reductions only increase slightly, and eventually yield no further benefit. At about 85% carbon footprint reduction, for cost limit of 1.6 and above, cost limit no longer shows any impacts since it is the performance of the CC retrofit and compensatory power that determine the reduction in emission level. Note that similar trends were also observed for Scenarios 1 to 3. The plants chosen to be retrofitted based on different cost limit for Base Case and Scenario 1 to 3 are summarised in Table S1–S4 in the supplementary file.

6. P-graph for carbon storage planning

Previous sections focus on choosing suitable carbon capture technologies for the power plants, with the main objective to maximise carbon footprint reduction, subject to a maximum increase of overall electricity cost. In this section, P-graph is applied to carbon storage planning, where storage capacity, geographical distance, and temporal constraints are considered. Two case studies are used to elucidate the use of P-graph model.

6.1. Case study 2

Case Study 2 is based on the results of Case Study 1 (Base Case). The main objective is to maximise CO_2 storage by determining optimum CO_2 source-sink matching under storage capacity and geographical distance constraints. Results of the base case scenario of Case Study 1 show that two CO_2 sources (Plants 1 and 2) are to be

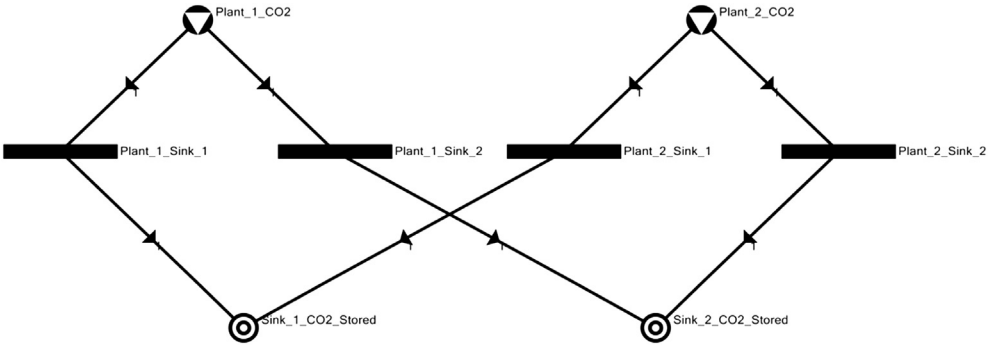


Fig. 6. P-graph representation for Case Study 2.

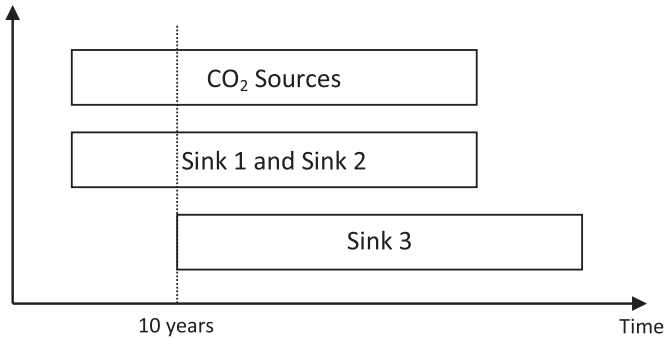


Fig. 7. Gantt chart representation of temporal issues in CO_2 source-sink matching.

Table 10
CO_2 source data for Case Study 3.

Sources	CO_2 flowrate (t/h)	Time of flow (y)	Maximum CO_2 capture (Mt)
Plant 1	540	0–46	200
Plant 2	450	0–41.5	150
Total			350

Table 11
CO_2 sink data for Case Study 3.

Sinks	Start time (y)	Maximum storage (Mt)
Sink 1	0	100
Sink 2	0	140
Sink 3	10	250
Total		490

Table 12
Distance between sources and sinks for Case Study 3.

Sources	Distance (km)		
	Sink 1	Sink 2	Sink 3
Plant 1	200	400	600
Plant 2	500	250	300

retrofitted with OFC, with streams given as in Table 7. In this case, it is assumed that only two geological reservoirs are available as CO_2 sinks and their characteristics are shown in Table 7. Furthermore, geographical distances between sources and sinks are taken into account in this case. Distances among source and sink are assumed but replicate real life situation (Table 8). It is assumed that the CO_2 transportation cost is directly proportional to distance.

Since the two sources generate a total of 350 Mt CO_2 throughout the planning period, while the combined storage capacity of the sinks is only 240 Mt, there is a CO_2 storage deficit of 110 Mt. Fig. 6 shows the P-graph representation for Case Study 2. The objective of this case study is to maximise CO_2 storage by matching carbon sources to the nearest sink. This also means that the solution will feature lowest transportation cost as the latter was assumed to be directly proportional to distance.

The resulting optimal allocation for this system is given in Table 9. The result shows that the storage capacity is insufficient such that only half of the CO_2 from Plant 1 can be stored in Sink 1 and a total of 140 Mt of CO_2 from Plant 2 is stored in Sink 2. The insufficiency of CO_2 storage capacity may be satisfied by discovering more CO_2 storage, or the captured CO_2 may be sent to external storage temporarily.

6.2. Case study 3

This case study makes use of the same data and carries the same objective as Case Study 2, except that an additional storage site (Sink 3) is now available. Thus, the storage will have enough capacity to start the total amount of CO_2 captured (350 Mt). Besides, the case study also considered the temporal aspect in source-sink matching, as the operating duration of sources and sinks may not completely overlap in reality. As illustrated in Fig. 7, Sink 3 begins to operate only after 10 years. The revised data are shown in Tables 10–12. Apart from that, the earliest time of availability of each sink is specified; however, it ceases to become viable when the storage capacity limit is reached. As shown in Table 11, the maximum capacity of the CO_2 Sink 3 is given as 250 Mt. Fig. 8 shows the P-graph representation for the overall structure of the problem for Case Study 3.

Solving the case study yields the optimal source-sink matching in Table 13. As shown, 43 Mt (21.5%) of CO_2 from Plant 1 and 36 Mt (24%) of CO_2 from Plant 2 are immediately matched to Sink 1 and

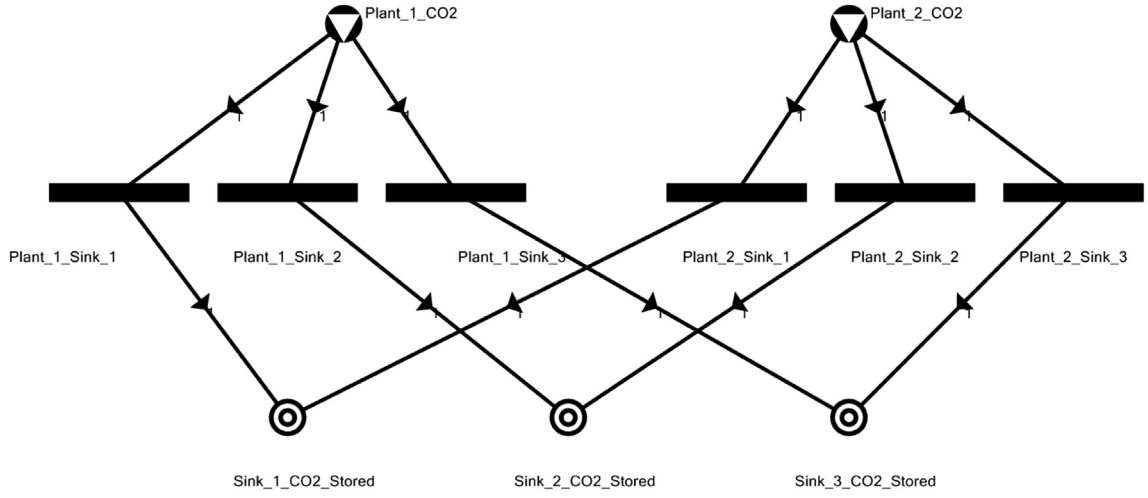


Fig. 8. P-graph representation for Case Study 3.

Table 13
Source-Sink Matching Matrix for Case Study 3 (values in parenthesis indicate year of operation).

Sources	Sink 1 (Mt)	Sink 2 (Mt)	Sink 3 (Mt)
Plant 1	43 (0–10) 53 (10–46)	104 (10–46)	
Plant 2		36 (0–10)	114 (10–41.5)
Surplus capacity	4		136

Sink 2 respectively from t = 0–10 years. In year 10, Plant 1 is connected to Sink 1 and Sink 2 simultaneously, while Plant 2 is connected to Sink 3 as soon as it is available. Based on Table 13, 96%, 100% and 45.6% of storage capacity of Sinks 1, 2 and 3 are utilised to store the captured CO_2, respectively.

7. Conclusions

The P-graph methodology for optimal planning of CCS deployment in power generation sector is demonstrated in this paper, and is demonstrated using realistic case studies from the literature. It takes into account of cost limits, carbon footprint reduction targets, power losses, selection of carbon capture technology and power plants for undergoing CCS retrofit, as well as the matching of capture CO_2 load with carbon storages. P-graph model shows globally optimal solutions that match those determined via MILP models. Variants of the basic problem are also solved, while trade-off and scenario analysis are performed to assess the interplay of various factors such as carbon footprint limits, cost limits, subsidies and technology performance levels. Future works should include unified model of CCS using P-graph [31] and extension of P-graph approach to account for geographical restrictions involving regional transfer of CO_2. Detailed modelling of CCS infrastructure (i.e., pipelines and pumping stations) [52] and sustainable network synthesis [53] may also be included.

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Nomenclatures

Indices

i	power plants (i = 1, 2, …, m)
j	CC technology (j = 1, 2, …, n)

Parameters

A_j	relative cost of electricity from plants retrofitted with CC technology (dimensionless)
B	relative cost of electricity from new plants to compensate for energy losses (dimensionless)
c	unit cost for power prior to retrofit ($/MW)
C_i	corresponding cost for power generated by power plants i ($/MW)
C_{ij}	corresponding cost for power related from power plants i to CC techniques j ($/MW)
C_j	corresponding cost for power output from CC techniques j ($/MW)
C_{total}	total cost corresponding to total power generated in the region ($/MW)
D	emission factor of power generation to compensate for CC energy losses (t CO_2/MWh)
L_j	relative energy loss associated with CC technology j (dimensionless)
m	number of power plants
n	number of CC techniques
P_i	power output targeted for power plants i (MWh)
P_{ij}	power related from power plants i to CC techniques j (MWh)
P_j	power output after passing through CC technique j (MWh)
P_{total}	total power output in the region (MWh)
RR_j	removal ratio of CC technique j (dimensionless)

Variables

x_{ij}	binary variable denoting decision to use CC technology j in plant i

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.energy.2014.08.060.

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